



## 1. Description

### 1.1. Project

|                 |                    |
|-----------------|--------------------|
| Project Name    | MotorDriverBoardv4 |
| Board Name      | custom             |
| Generated with: | STM32CubeMX 6.15.0 |
| Date            | 03/13/2026         |

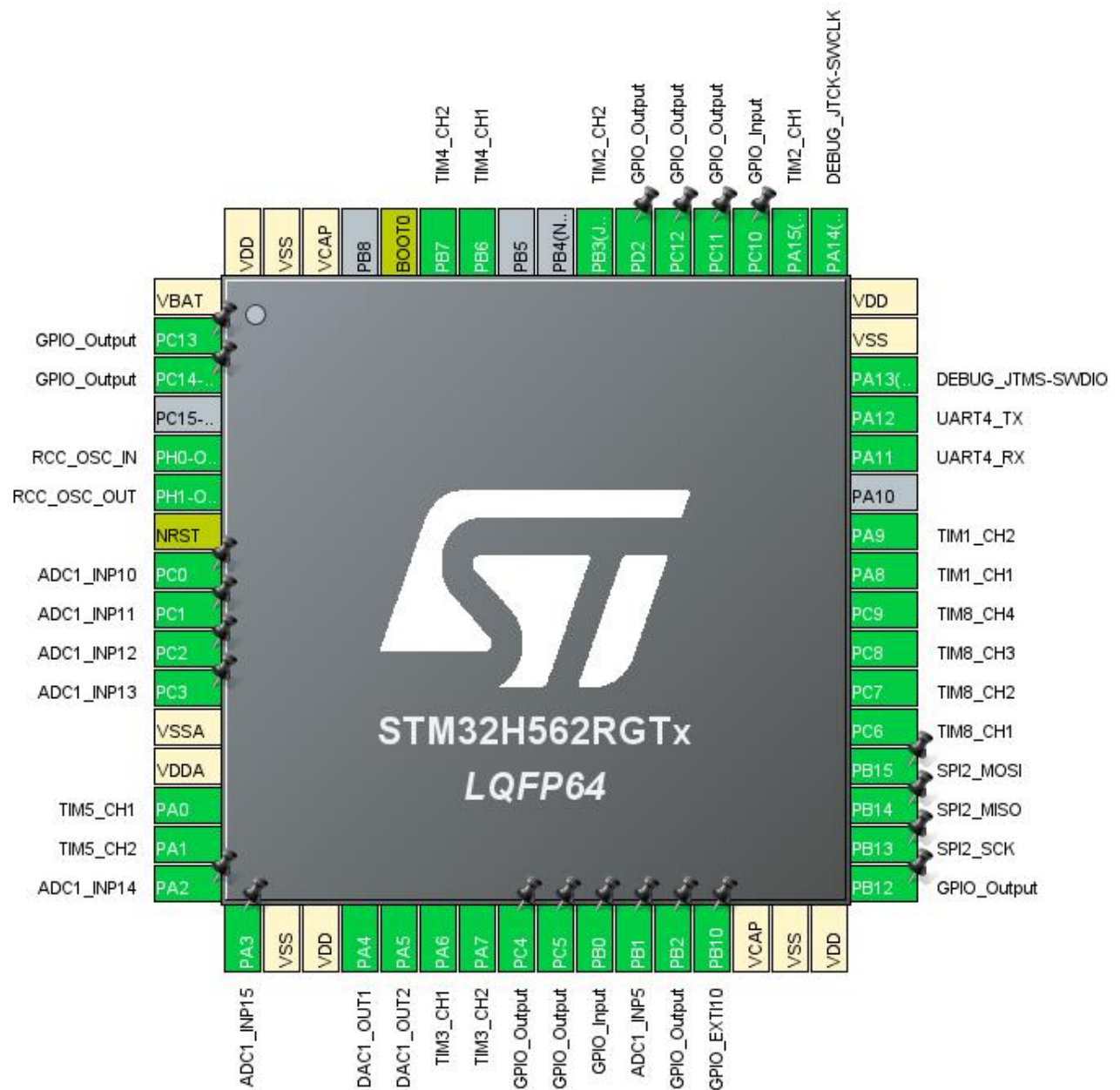
### 1.2. MCU

|                |               |
|----------------|---------------|
| MCU Series     | STM32H5       |
| MCU Line       | STM32H5x2     |
| MCU name       | STM32H562RGTx |
| MCU Package    | LQFP64        |
| MCU Pin number | 64            |

### 1.3. Core(s) information

|         |                |
|---------|----------------|
| Core(s) | ARM Cortex-M33 |
|---------|----------------|

## 2. Pinout Configuration



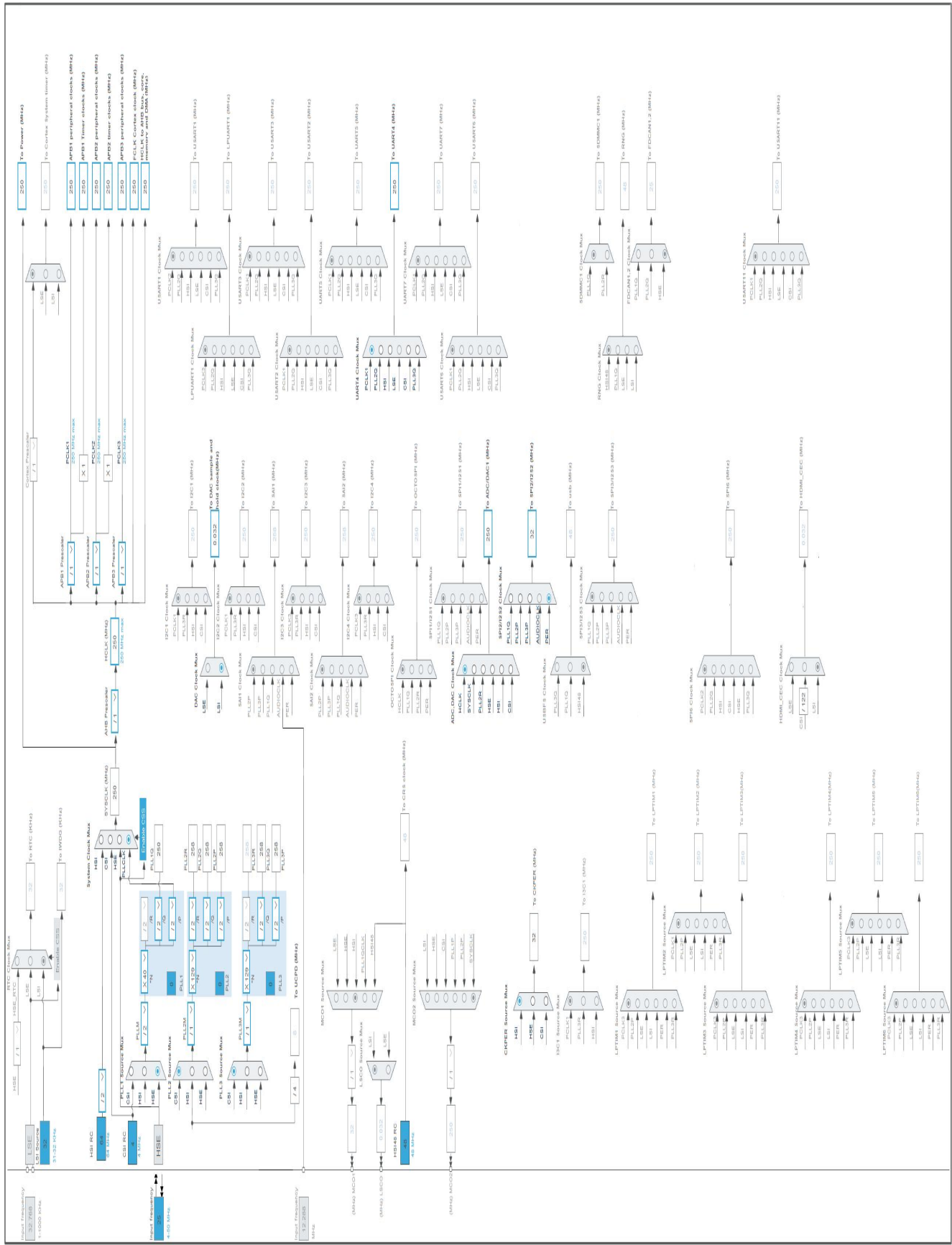
### 3. Pins Configuration

| Pin Number<br>LQFP64 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label |
|----------------------|---------------------------------------|----------|--------------------------|-------|
| 1                    | VBAT                                  | Power    |                          |       |
| 2                    | PC13 *                                | I/O      | GPIO_Output              |       |
| 3                    | PC14-<br>OSC32_IN(OSC32_IN) *         | I/O      | GPIO_Output              |       |
| 5                    | PH0-OSC_IN(PH0)                       | I/O      | RCC_OSC_IN               |       |
| 6                    | PH1-OSC_OUT(PH1)                      | I/O      | RCC_OSC_OUT              |       |
| 7                    | NRST                                  | Reset    |                          |       |
| 8                    | PC0                                   | I/O      | ADC1_INP10               |       |
| 9                    | PC1                                   | I/O      | ADC1_INP11               |       |
| 10                   | PC2                                   | I/O      | ADC1_INP12               |       |
| 11                   | PC3                                   | I/O      | ADC1_INP13               |       |
| 12                   | VSSA                                  | Power    |                          |       |
| 13                   | VDDA                                  | Power    |                          |       |
| 14                   | PA0                                   | I/O      | TIM5_CH1                 |       |
| 15                   | PA1                                   | I/O      | TIM5_CH2                 |       |
| 16                   | PA2                                   | I/O      | ADC1_INP14               |       |
| 17                   | PA3                                   | I/O      | ADC1_INP15               |       |
| 18                   | VSS                                   | Power    |                          |       |
| 19                   | VDD                                   | Power    |                          |       |
| 20                   | PA4                                   | I/O      | DAC1_OUT1                |       |
| 21                   | PA5                                   | I/O      | DAC1_OUT2                |       |
| 22                   | PA6                                   | I/O      | TIM3_CH1                 |       |
| 23                   | PA7                                   | I/O      | TIM3_CH2                 |       |
| 24                   | PC4 *                                 | I/O      | GPIO_Output              |       |
| 25                   | PC5 *                                 | I/O      | GPIO_Output              |       |
| 26                   | PB0 *                                 | I/O      | GPIO_Input               |       |
| 27                   | PB1                                   | I/O      | ADC1_INP5                |       |
| 28                   | PB2 *                                 | I/O      | GPIO_Output              |       |
| 29                   | PB10                                  | I/O      | GPIO_EXTI10              |       |
| 30                   | VCAP                                  | Power    |                          |       |
| 31                   | VSS                                   | Power    |                          |       |
| 32                   | VDD                                   | Power    |                          |       |
| 33                   | PB12 *                                | I/O      | GPIO_Output              |       |
| 34                   | PB13                                  | I/O      | SPI2_SCK                 |       |
| 35                   | PB14                                  | I/O      | SPI2_MISO                |       |
| 36                   | PB15                                  | I/O      | SPI2_MOSI                |       |

| Pin Number<br>LQFP64 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label |
|----------------------|---------------------------------------|----------|--------------------------|-------|
| 37                   | PC6                                   | I/O      | TIM8_CH1                 |       |
| 38                   | PC7                                   | I/O      | TIM8_CH2                 |       |
| 39                   | PC8                                   | I/O      | TIM8_CH3                 |       |
| 40                   | PC9                                   | I/O      | TIM8_CH4                 |       |
| 41                   | PA8                                   | I/O      | TIM1_CH1                 |       |
| 42                   | PA9                                   | I/O      | TIM1_CH2                 |       |
| 44                   | PA11                                  | I/O      | UART4_RX                 |       |
| 45                   | PA12                                  | I/O      | UART4_TX                 |       |
| 46                   | PA13(JTMS/SWDIO)                      | I/O      | DEBUG_JTMS-SWDIO         |       |
| 47                   | VSS                                   | Power    |                          |       |
| 48                   | VDD                                   | Power    |                          |       |
| 49                   | PA14(JTCK/SWCLK)                      | I/O      | DEBUG_JTCK-SWCLK         |       |
| 50                   | PA15(JTDI)                            | I/O      | TIM2_CH1                 |       |
| 51                   | PC10 *                                | I/O      | GPIO_Input               |       |
| 52                   | PC11 *                                | I/O      | GPIO_Output              |       |
| 53                   | PC12 *                                | I/O      | GPIO_Output              |       |
| 54                   | PD2 *                                 | I/O      | GPIO_Output              |       |
| 55                   | PB3(JTDO/TRACESWO)                    | I/O      | TIM2_CH2                 |       |
| 58                   | PB6                                   | I/O      | TIM4_CH1                 |       |
| 59                   | PB7                                   | I/O      | TIM4_CH2                 |       |
| 60                   | BOOT0                                 | Boot     |                          |       |
| 62                   | VCAP                                  | Power    |                          |       |
| 63                   | VSS                                   | Power    |                          |       |
| 64                   | VDD                                   | Power    |                          |       |

\* The pin is affected with an I/O function

## 4. Clock Tree Configuration



## 1. Power Consumption Calculator report

### 1.1. Microcontroller Selection

|           |               |
|-----------|---------------|
| Series    | STM32H5       |
| Line      | STM32H5x2     |
| MCU       | STM32H562RGTx |
| Datasheet | DS00000 Rev0  |

### 1.2. Parameter Selection

|             |     |
|-------------|-----|
| Temperature | 25  |
| Vdd         | 3.0 |

### 1.3. Battery Selection

|                   |                 |
|-------------------|-----------------|
| Battery           | Li-SOCL2(A3400) |
| Capacity          | 3400.0 mAh      |
| Self Discharge    | 0.08 %/month    |
| Nominal Voltage   | 3.6 V           |
| Max Cont Current  | 100.0 mA        |
| Max Pulse Current | 200.0 mA        |
| Cells in series   | 1               |
| Cells in parallel | 1               |

## 1.4. Sequence

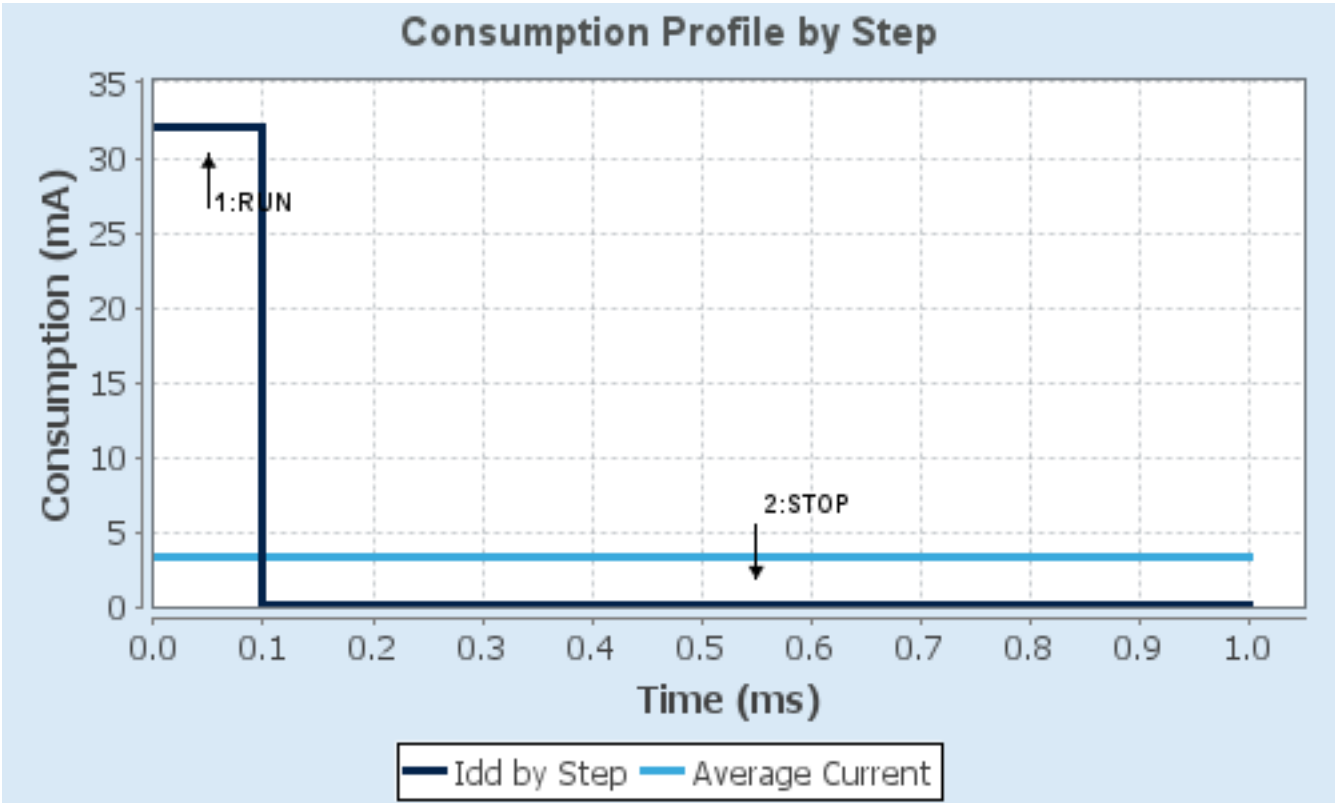
|                               |                                           |                                 |
|-------------------------------|-------------------------------------------|---------------------------------|
| <b>Step</b>                   | Step1                                     | Step2                           |
| <b>Mode</b>                   | RUN                                       | STOP                            |
| <b>Vdd</b>                    | 3.0                                       | 3.0                             |
| <b>Voltage Source</b>         | Battery                                   | Battery                         |
| <b>Range</b>                  | VOS0: Scale0                              | SVOS5: System-Scale5/SMPS       |
| <b>Fetch Type</b>             | FLASH_ON/Cache2Ways_A<br>LL_RAM_RETENTION | Flash-<br>PwrDwn_PwrDwnStop_OFF |
| <b>CPU Frequency</b>          | 250 MHz                                   | 0 Hz                            |
| <b>Clock Configuration</b>    | HSE BYP PLL                               | ALL_CLOCKS_OFF                  |
| <b>Clock Source Frequency</b> | 8 MHz                                     | 0 Hz                            |
| <b>Peripherals</b>            |                                           |                                 |
| <b>Additional Cons.</b>       | 0 mA                                      | 0 mA                            |
| <b>Average Current</b>        | 32 mA                                     | 51.5 $\mu$ A                    |
| <b>Duration</b>               | 0.1 ms                                    | 0.9 ms                          |
| <b>DMIPS</b>                  | 535.0                                     | 0.0                             |
| <b>Ta Max</b>                 | 120.68                                    | 124.99                          |
| <b>Category</b>               | In DS Table                               | In DS Table                     |

## 1.5. Results

|               |                              |                 |             |
|---------------|------------------------------|-----------------|-------------|
| Sequence Time | 1 ms                         | Average Current | 3.25 mA     |
| Battery Life  | 1 month, 13 days,<br>4 hours | Average DMIPS   | 535.0 DMIPS |

## 1.6. Chart





## 2. Software Project

### 2.1. Project Settings

| Name                              | Value                                                                   |
|-----------------------------------|-------------------------------------------------------------------------|
| Project Name                      | MotorDriverBoardv4                                                      |
| Project Folder                    | C:\Users\Michael\STM32CubeIDE\workspace_1.19.0\MotorDriverBoardv5\Cubel |
| Toolchain / IDE                   | STM32CubeIDE                                                            |
| Firmware Package Name and Version | STM32Cube FW_H5 V1.5.1                                                  |
| Application Structure             | Advanced                                                                |
| Generate Under Root               | Yes                                                                     |
| Do not generate the main()        | No                                                                      |
| Minimum Heap Size                 | 0x200                                                                   |
| Minimum Stack Size                | 0x400                                                                   |

### 2.2. Code Generation Settings

| Name                                                            | Value                                 |
|-----------------------------------------------------------------|---------------------------------------|
| STM32Cube MCU packages and embedded software                    | Copy only the necessary library files |
| Generate peripheral initialization as a pair of '.c/.h' files   | No                                    |
| Backup previously generated files when re-generating            | No                                    |
| Keep User Code when re-generating                               | Yes                                   |
| Delete previously generated files when not re-generated         | Yes                                   |
| Set all free pins as analog (to optimize the power consumption) | No                                    |
| Enable Full Assert                                              | No                                    |

### 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name      | Peripheral Instance Name |
|------|--------------------|--------------------------|
| 1    | SystemClock_Config | RCC                      |
| 2    | MX_GPIO_Init       | GPIO                     |
| 3    | MX_GPDMA1_Init     | GPDMA1                   |
| 4    | MX_ADC1_Init       | ADC1                     |
| 5    | MX_DAC1_Init       | DAC1                     |
| 6    | MX_SPI2_Init       | SPI2                     |
| 7    | MX_TIM1_Init       | TIM1                     |
| 8    | MX_TIM2_Init       | TIM2                     |
| 9    | MX_TIM3_Init       | TIM3                     |
| 10   | MX_TIM4_Init       | TIM4                     |
| 11   | MX_TIM5_Init       | TIM5                     |

| Rank | Function Name  | Peripheral Instance Name |
|------|----------------|--------------------------|
| 12   | MX_TIM8_Init   | TIM8                     |
| 13   | MX_ICACHE_Init | ICACHE                   |
| 14   | MX_UART4_Init  | UART4                    |

### 3. Peripherals and Middlewares Configuration

#### 3.1. ADC1

IN5: IN5 Single-ended

IN10: IN10 Single-ended

IN11: IN11 Single-ended

IN12: IN12 Single-ended

mode: IN13

mode: IN14

mode: IN15

##### 3.1.1. Parameter Settings:

##### ADCs\_Common\_Settings:

Mode Independent mode

##### ADC\_Settings:

Clock Prescaler Asynchronous clock mode divided by 4

Resolution ADC 12-bit resolution

Scan Conversion Mode Enabled

Data Alignment Right alignment

Continuous Conversion Mode **Enabled \***

Discontinuous Conversion Mode Disabled

DMA Continuous Requests **Enabled \***

End Of Conversion Selection End of single conversion

Overrun behaviour **Overrun data overwritten \***

Conversion Data Management Mode Regular Conversion data stored in DR register only

Low Power Auto Wait Disabled

##### ADC\_Regular\_ConversionMode:

Enable Regular Conversions Enable

Enable Regular Oversampling Disable

Number Of Conversion **7 \***

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Sampling Mode Normal

Rank 1

Channel **Channel 14 \***

Sampling Time **92.5 Cycles \***

Offset Number No offset

Monitored by None

Rank **2 \***

|                                     |                      |
|-------------------------------------|----------------------|
| Channel                             | <b>Channel 15 *</b>  |
| Sampling Time                       | <b>92.5 Cycles *</b> |
| Offset Number                       | No offset            |
| Monitored by                        | None                 |
| <u>Rank</u>                         | <b>3 *</b>           |
| Channel                             | <b>Channel 11 *</b>  |
| Sampling Time                       | <b>92.5 Cycles *</b> |
| Offset Number                       | No offset            |
| Monitored by                        | None                 |
| <u>Rank</u>                         | <b>4 *</b>           |
| Channel                             | <b>Channel 10 *</b>  |
| Sampling Time                       | <b>92.5 Cycles *</b> |
| Offset Number                       | No offset            |
| Monitored by                        | None                 |
| <u>Rank</u>                         | <b>5 *</b>           |
| Channel                             | <b>Channel 12 *</b>  |
| Sampling Time                       | <b>92.5 Cycles *</b> |
| Offset Number                       | No offset            |
| Monitored by                        | None                 |
| <u>Rank</u>                         | <b>6 *</b>           |
| Channel                             | <b>Channel 13 *</b>  |
| Sampling Time                       | <b>92.5 Cycles *</b> |
| Offset Number                       | No offset            |
| Monitored by                        | None                 |
| <u>Rank</u>                         | <b>7 *</b>           |
| Channel                             | Channel 5            |
| Sampling Time                       | <b>92.5 Cycles *</b> |
| Offset Number                       | No offset            |
| Monitored by                        | None                 |
| <b>ADC_Injected_ConversionMode:</b> |                      |
| Enable Injected Conversions         | Disable              |
| <b>Analog Watchdog 1:</b>           |                      |
| Enable Analog WatchDog1 Mode        | false                |
| <b>Analog Watchdog 2:</b>           |                      |
| Enable Analog WatchDog2 Mode        | false                |
| <b>Analog Watchdog 3:</b>           |                      |
| Enable Analog WatchDog3 Mode        | false                |

### 3.2. BOOTPATH

mode: Activated

### 3.3. DAC1

OUT1 connected to: only external pin

OUT2 connected to: only external pin

#### 3.3.1. Parameter Settings:

##### Common DAC Settings:

|                         |         |
|-------------------------|---------|
| DAC High Frequency Mode | Disable |
|-------------------------|---------|

##### DAC Out1 Settings:

|                      |                  |
|----------------------|------------------|
| Mode selected        | Normal Mode      |
| Output Buffer        | Enable           |
| DMA double data mode | Disable          |
| Signed Format        | Disable          |
| Trigger              | None             |
| User Trimming        | Factory trimming |

##### DAC Out2 Settings:

|                      |                  |
|----------------------|------------------|
| Mode selected        | Normal Mode      |
| Output Buffer        | Enable           |
| DMA double data mode | Disable          |
| Signed Format        | Disable          |
| Trigger              | None             |
| User Trimming        | Factory trimming |

### 3.4. DEBUG

Debug: Serial Wire

### 3.5. ICACHE

Mode: 2-ways set associative cache

### 3.6. MEMORYMAP

mode: Activated

### 3.7. PWR

**mode: Power saving mode**

**mode: Privilege attributes**

3.7.1. PWR Privilege :

**Privilege PWR:**

|               |         |
|---------------|---------|
| PWR Privilege | Disable |
|---------------|---------|

**3.8. RCC**

**High Speed Clock (HSE): Crystal/Ceramic Resonator**

3.8.1. RCC Privilege :

**Privilege RCC:**

|                                   |         |
|-----------------------------------|---------|
| Privilege of RCC Non-Secure Items | Disable |
|-----------------------------------|---------|

3.8.2. Parameter Settings:

**System Parameters:**

|                         |                    |
|-------------------------|--------------------|
| VDD voltage (V)         | 3.3                |
| Flash Latency(WS)       | 5 WS (6 CPU cycle) |
| Flash Programming Delay | 2                  |

**RCC Parameters:**

|                                |          |
|--------------------------------|----------|
| HSI Calibration Value          | 64       |
| CSI Calibration Value          | 32       |
| HSE Startup Timeout Value (ms) | 100      |
| LSE Startup Timeout Value (ms) | 5000     |
| TIM Prescaler Selection        | Disabled |

**Power Parameters:**

|                               |                                 |
|-------------------------------|---------------------------------|
| Power Regulator Voltage Scale | Power Regulator Voltage Scale 0 |
|-------------------------------|---------------------------------|

**3.9. SPI2**

**Mode: Full-Duplex Master**

3.9.1. Parameter Settings:

**Basic Parameters:**

|              |          |
|--------------|----------|
| Frame Format | Motorola |
| Data Size    | 4 Bits   |

|                             |                              |
|-----------------------------|------------------------------|
| First Bit                   | MSB First                    |
| <b>Clock Parameters:</b>    |                              |
| Prescaler (for Baud Rate)   | 2                            |
| Baud Rate                   | <b>16.0 MBits/s *</b>        |
| Clock Polarity (CPOL)       | Low                          |
| Clock Phase (CPHA)          | 1 Edge                       |
| <b>CRC Parameters:</b>      |                              |
| CRC Calculation             | Disabled                     |
| <b>Advanced Parameters:</b> |                              |
| NSSP Mode                   | Enabled                      |
| NSS Signal Type             | Software                     |
| Fifo Threshold              | Fifo Threshold 01 Data       |
| Nss Polarity                | Nss Polarity Low             |
| Master Ss Idleness          | 00 Cycle                     |
| Master Inter Data Idleness  | 00 Cycle                     |
| Master Receiver Auto Susp   | Disable                      |
| Master Keep Io State        | Master Keep Io State Disable |
| IO Swap                     | Disabled                     |
| Ready Master Management     | Internal                     |
| Ready Signal Polarity       | High                         |

### 3.10. SYS

**Timebase Source: TIM17**

### 3.11. TIM1

**Channel1: PWM Generation CH1**

**Channel2: PWM Generation CH2**

#### 3.11.1. Parameter Settings:

##### **Counter Settings:**

|                                                       |                 |
|-------------------------------------------------------|-----------------|
| Prescaler (PSC - 16 bits value)                       | 0               |
| Counter Mode                                          | Up              |
| Dithering                                             | Disable         |
| Counter Period (AutoReload Register - 16 bits value ) | <b>9999 *</b>   |
| Internal Clock Division (CKD)                         | No Division     |
| Repetition Counter (RCR - 16 bits value)              | 0               |
| auto-reload preload                                   | <b>Enable *</b> |

##### **Trigger Output (TRGO) Parameters:**



|                               |                                            |
|-------------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit)   | Disable (Trigger input effect not delayed) |
| Trigger Event Selection TRGO  | Reset (UG bit from TIMx_EGR)               |
| Trigger Event Selection TRGO2 | Reset (UG bit from TIMx_EGR)               |

#### Break And Dead Time management - BRK Configuration:

|                           |         |
|---------------------------|---------|
| BRK State                 | Disable |
| BRK Polarity              | High    |
| BRK Filter (4 bits value) | 0       |
| BRK Sources Configuration |         |
| - Digital Input           | Disable |

#### Break And Dead Time management - BRK2 Configuration:

|                            |         |
|----------------------------|---------|
| BRK2 State                 | Disable |
| BRK2 Polarity              | High    |
| BRK2 Filter (4 bits value) | 0       |
| BRK2 Sources Configuration |         |
| - Digital Input            | Disable |

#### Break And Dead Time management - Output Configuration:

|                                          |         |
|------------------------------------------|---------|
| Automatic Output State                   | Disable |
| Off State Selection for Run Mode (OSSR)  | Disable |
| Off State Selection for Idle Mode (OSSI) | Disable |
| Lock Configuration                       | Off     |

#### Clear Input:

|                    |         |
|--------------------|---------|
| Clear Input Source | Disable |
|--------------------|---------|

#### PWM Generation Channel 1:

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (16 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |
| CH Idle State          | Reset      |

#### PWM Generation Channel 2:

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (16 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |
| CH Idle State          | Reset      |

### 3.12. TIM2

#### Combined Channels: Encoder Mode

### 3.12.1. Parameter Settings:

#### **Counter Settings:**

|                                                       |             |
|-------------------------------------------------------|-------------|
| Prescaler (PSC - 16 bits value)                       | 0           |
| Counter Mode                                          | Up          |
| Dithering                                             | Disable     |
| Counter Period (AutoReload Register - 32 bits value ) | 4294967295  |
| Internal Clock Division (CKD)                         | No Division |
| auto-reload preload                                   | Disable     |

#### **Trigger Output (TRGO) Parameters:**

|                              |                                            |
|------------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit)  | Disable (Trigger input effect not delayed) |
| Trigger Event Selection TRGO | Reset (UG bit from TIMx_EGR)               |

#### **Encoder:**

|                                    |                  |
|------------------------------------|------------------|
| Encoder Mode                       | Encoder Mode T11 |
| Slave Mode Preload Activation      | Disable          |
| ____ Parameters for Channel 1 ____ |                  |
| Polarity                           | Rising Edge      |
| IC Selection                       | Direct           |
| Prescaler Division Ratio           | No division      |
| Input Filter                       | 0                |
| ____ Parameters for Channel 2 ____ |                  |
| Polarity                           | Rising Edge      |
| IC Selection                       | Direct           |
| Prescaler Division Ratio           | No division      |
| Input Filter                       | 0                |

## **3.13. TIM3**

### **Combined Channels: Encoder Mode**

#### 3.13.1. Parameter Settings:

#### **Counter Settings:**

|                                                       |             |
|-------------------------------------------------------|-------------|
| Prescaler (PSC - 16 bits value)                       | 0           |
| Counter Mode                                          | Up          |
| Dithering                                             | Disable     |
| Counter Period (AutoReload Register - 16 bits value ) | 65535       |
| Internal Clock Division (CKD)                         | No Division |
| auto-reload preload                                   | Disable     |

#### **Trigger Output (TRGO) Parameters:**

|                             |                                            |
|-----------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit) | Disable (Trigger input effect not delayed) |
|-----------------------------|--------------------------------------------|

Trigger Event Selection TRGO Reset (UG bit from TIMx\_EGR)

**Encoder:**

Encoder Mode Encoder Mode T11

Slave Mode Preload Activation Disable

\_\_\_\_ Parameters for Channel 1 \_\_\_\_

Polarity Rising Edge

IC Selection Direct

Prescaler Division Ratio No division

Input Filter 0

\_\_\_\_ Parameters for Channel 2 \_\_\_\_

Polarity Rising Edge

IC Selection Direct

Prescaler Division Ratio No division

Input Filter 0

### 3.14. TIM4

#### Combined Channels: Encoder Mode

##### 3.14.1. Parameter Settings:

**Counter Settings:**

Prescaler (PSC - 16 bits value) 0

Counter Mode Up

Dithering Disable

Counter Period (AutoReload Register - 16 bits value ) 65535

Internal Clock Division (CKD) No Division

auto-reload preload Disable

**Trigger Output (TRGO) Parameters:**

Master/Slave Mode (MSM bit) Disable (Trigger input effect not delayed)

Trigger Event Selection TRGO Reset (UG bit from TIMx\_EGR)

**Encoder:**

Encoder Mode Encoder Mode T11

Slave Mode Preload Activation Disable

\_\_\_\_ Parameters for Channel 1 \_\_\_\_

Polarity Rising Edge

IC Selection Direct

Prescaler Division Ratio No division

Input Filter 0

\_\_\_\_ Parameters for Channel 2 \_\_\_\_

Polarity Rising Edge

IC Selection Direct

|                          |             |
|--------------------------|-------------|
| Prescaler Division Ratio | No division |
| Input Filter             | 0           |

### 3.15. TIM5

#### Combined Channels: Encoder Mode

##### 3.15.1. Parameter Settings:

###### Counter Settings:

|                                                       |             |
|-------------------------------------------------------|-------------|
| Prescaler (PSC - 16 bits value)                       | 0           |
| Counter Mode                                          | Up          |
| Dithering                                             | Disable     |
| Counter Period (AutoReload Register - 32 bits value ) | 4294967295  |
| Internal Clock Division (CKD)                         | No Division |
| auto-reload preload                                   | Disable     |

###### Trigger Output (TRGO) Parameters:

|                              |                                            |
|------------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit)  | Disable (Trigger input effect not delayed) |
| Trigger Event Selection TRGO | Reset (UG bit from TIMx_EGR)               |

###### Encoder:

|                                    |                  |
|------------------------------------|------------------|
| Encoder Mode                       | Encoder Mode T11 |
| Slave Mode Preload Activation      | Disable          |
| ____ Parameters for Channel 1 ____ |                  |
| Polarity                           | Rising Edge      |
| IC Selection                       | Direct           |
| Prescaler Division Ratio           | No division      |
| Input Filter                       | 0                |
| ____ Parameters for Channel 2 ____ |                  |
| Polarity                           | Rising Edge      |
| IC Selection                       | Direct           |
| Prescaler Division Ratio           | No division      |
| Input Filter                       | 0                |

### 3.16. TIM8

**Channel1: PWM Generation CH1**

**Channel2: PWM Generation CH2**

**Channel3: PWM Generation CH3**

**Channel4: PWM Generation CH4**

### 3.16.1. Parameter Settings:

#### Counter Settings:

|                                                       |                 |
|-------------------------------------------------------|-----------------|
| Prescaler (PSC - 16 bits value)                       | 0               |
| Counter Mode                                          | Up              |
| Dithering                                             | Disable         |
| Counter Period (AutoReload Register - 16 bits value ) | <b>9999 *</b>   |
| Internal Clock Division (CKD)                         | No Division     |
| Repetition Counter (RCR - 16 bits value)              | 0               |
| auto-reload preload                                   | <b>Enable *</b> |

#### Trigger Output (TRGO) Parameters:

|                               |                                            |
|-------------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit)   | Disable (Trigger input effect not delayed) |
| Trigger Event Selection TRGO  | Reset (UG bit from TIMx_EGR)               |
| Trigger Event Selection TRGO2 | Reset (UG bit from TIMx_EGR)               |

#### Break And Dead Time management - BRK Configuration:

|                           |         |
|---------------------------|---------|
| BRK State                 | Disable |
| BRK Polarity              | High    |
| BRK Filter (4 bits value) | 0       |
| BRK Sources Configuration |         |
| - Digital Input           | Disable |

#### Break And Dead Time management - BRK2 Configuration:

|                            |         |
|----------------------------|---------|
| BRK2 State                 | Disable |
| BRK2 Polarity              | High    |
| BRK2 Filter (4 bits value) | 0       |
| BRK2 Sources Configuration |         |
| - Digital Input            | Disable |

#### Break And Dead Time management - Output Configuration:

|                                          |         |
|------------------------------------------|---------|
| Automatic Output State                   | Disable |
| Off State Selection for Run Mode (OSSR)  | Disable |
| Off State Selection for Idle Mode (OSSI) | Disable |
| Lock Configuration                       | Off     |

#### Clear Input:

|                    |         |
|--------------------|---------|
| Clear Input Source | Disable |
|--------------------|---------|

#### PWM Generation Channel 1:

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (16 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |
| CH Idle State          | Reset      |

#### PWM Generation Channel 2:

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (16 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |
| CH Idle State          | Reset      |

#### **PWM Generation Channel 3:**

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (16 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |
| CH Idle State          | Reset      |

#### **PWM Generation Channel 4:**

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (16 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |
| CH Idle State          | Reset      |

### **3.17. UART4**

#### **Mode: Asynchronous**

##### 3.17.1. Parameter Settings:

#### **Basic Parameters:**

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

#### **Advanced Parameters:**

|                  |                             |
|------------------|-----------------------------|
| Data Direction   | Receive and Transmit        |
| Over Sampling    | 16 Samples                  |
| Single Sample    | Disable                     |
| ClockPrescaler   | 1                           |
| Fifo Mode        | FIFO mode disable           |
| Txfifo Threshold | 1 eighth full configuration |
| Rxfifo Threshold | 1 eighth full configuration |

#### **Advanced Features:**

|               |         |
|---------------|---------|
| Auto Baudrate | Disable |
|---------------|---------|

|                               |         |
|-------------------------------|---------|
| TX Pin Active Level Inversion | Disable |
| RX Pin Active Level Inversion | Disable |
| Data Inversion                | Disable |
| TX and RX Pins Swapping       | Disable |
| Overrun                       | Enable  |
| DMA on RX Error               | Enable  |
| MSB First                     | Disable |

### 3.18. STMicroelectronics.X-CUBE-FREERTOS.1.3.1

#### mode: CMSISJjRTOS2

##### 3.18.1. Config parameters:

###### API:

|              |             |
|--------------|-------------|
| FreeRTOS API | CMSIS-RTOS2 |
|--------------|-------------|

###### Versions:

|                    |        |
|--------------------|--------|
| FreeRTOS version   | 10.6.2 |
| CMSIS-RTOS version | 2.1.3  |

###### Security settings:

|                          |          |
|--------------------------|----------|
| ENABLE_TRUSTZONE         | Disabled |
| RUN_FREERTOS_SECURE_ONLY | Disabled |

###### MPU/FPU:

|            |          |
|------------|----------|
| ENABLE_FPU | Disabled |
|------------|----------|

###### MVE:

|                  |          |
|------------------|----------|
| configENABLE_MVE | Disabled |
|------------------|----------|

###### Kernel settings:

|                                   |                 |
|-----------------------------------|-----------------|
| USE_PREEMPTION                    | Enabled         |
| CPU_CLOCK_HZ                      | SystemCoreClock |
| TICK_RATE_HZ                      | 1000            |
| MAX_PRIORITIES                    | 56              |
| MINIMAL_STACK_SIZE                | 128             |
| MAX_TASK_NAME_LEN                 | 16              |
| USE_16_BIT_TICKS                  | Disabled        |
| IDLE_SHOULD_YIELD                 | Enabled         |
| USE_MUTEXES                       | Enabled         |
| USE_RECURSIVE_MUTEXES             | Enabled         |
| USE_COUNTING_SEMAPHORES           | Enabled         |
| QUEUE_REGISTRY_SIZE               | 8               |
| USE_APPLICATION_TASK_TAG          | Disabled        |
| ENABLE_BACKWARD_COMPATIBILITY     | Enabled         |
| USE_PORT_OPTIMISED_TASK_SELECTION | Disabled        |

|                           |          |
|---------------------------|----------|
| USE_TICKLESS_IDLE         | Disabled |
| USE_TASK_NOTIFICATIONS    | Enabled  |
| RECORD_STACK_HIGH_ADDRESS | Disabled |
| USE_MINI_LIST_ITEM        | Enabled  |
| USE_SB_COMPLETED_CALLBACK | Disabled |

### Memory management settings:

|                                     |                  |
|-------------------------------------|------------------|
| Memory Allocation                   | Dynamic / Static |
| TOTAL_HEAP_SIZE                     | 8192             |
| STACK_ALLOCATION_FROM_SEPARATE_HEAP | Disabled         |
| HEAP_CLEAR_MEMORY_ON_FREE           | Disabled         |

### Hook function related definitions:

|                              |          |
|------------------------------|----------|
| USE_IDLE_HOOK                | Disabled |
| USE_TICK_HOOK                | Disabled |
| USE_MALLOC_FAILED_HOOK       | Disabled |
| USE_DAEMON_TASK_STARTUP_HOOK | Disabled |
| CHECK_FOR_STACK_OVERFLOW     | Disabled |

### Run time and task stats gathering related definitions:

|                                |          |
|--------------------------------|----------|
| GENERATE_RUN_TIME_STATS        | Disabled |
| USE_TRACE_FACILITY             | Enabled  |
| USE_STATS_FORMATTING_FUNCTIONS | Disabled |
| RUN_TIME_COUNTER_TYPE          | uint32_t |

### Co-routine related definitions:

|                           |          |
|---------------------------|----------|
| USE_CO_ROUTINES           | Disabled |
| MAX_CO_ROUTINE_PRIORITIES | 2        |

### Software timer definitions:

|                        |         |
|------------------------|---------|
| USE_TIMERS             | Enabled |
| TIMER_TASK_PRIORITY    | 2       |
| TIMER_QUEUE_LENGTH     | 10      |
| TIMER_TASK_STACK_DEPTH | 128     |

### Interrupt nesting behaviour configuration:

|                                        |    |
|----------------------------------------|----|
| LIBRARY_LOWEST_INTERRUPT_PRIORITY      | 15 |
| LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY | 5  |

### Added with 10.2.1 support:

|                            |          |
|----------------------------|----------|
| MESSAGE_BUFFER_LENGTH_TYPE | size_t   |
| USE_POSIX_ERRNO            | Disabled |

### CMSIS-RTOS V2 flags:

|                               |         |
|-------------------------------|---------|
| USE_OS2_THREAD_SUSPEND_RESUME | Enabled |
| USE_OS2_THREAD_ENUMERATE      | Enabled |
| USE_OS2_EVENTFLAGS_FROM_ISR   | Enabled |
| USE_OS2_THREAD_FLAGS          | Enabled |
| USE_OS2_TIMER                 | Enabled |



|               |         |
|---------------|---------|
| USE_OS2_MUTEX | Enabled |
|---------------|---------|

### 3.18.2. Include parameters:

#### **Include definitions:**

|                              |          |
|------------------------------|----------|
| vTaskPrioritySet             | Enabled  |
| uxTaskPriorityGet            | Enabled  |
| vTaskDelete                  | Enabled  |
| vTaskCleanUpResources        | Disabled |
| vTaskSuspend                 | Enabled  |
| xTaskDelayUntil              | Enabled  |
| vTaskDelay                   | Enabled  |
| xTaskGetSchedulerState       | Enabled  |
| xTaskResumeFromISR           | Enabled  |
| xQueueGetMutexHolder         | Enabled  |
| xSemaphoreGetMutexHolder     | Enabled  |
| pcTaskGetTaskName            | Disabled |
| uxTaskGetStackHighWaterMark  | Enabled  |
| xTaskGetCurrentTaskHandle    | Enabled  |
| eTaskGetState                | Enabled  |
| xEventGroupSetBitFromISR     | Disabled |
| xTimerPendFunctionCall       | Enabled  |
| xTaskAbortDelay              | Disabled |
| xTaskGetHandle               | Disabled |
| uxTaskGetStackHighWaterMark2 | Disabled |

### 3.18.3. Advanced settings:

#### **Newlib settings (see parameter description first):**

|                      |          |
|----------------------|----------|
| USE_NEWLIB_REENTRANT | Disabled |
|----------------------|----------|

#### **Project settings (see parameter description first):**

|                    |         |
|--------------------|---------|
| Use pack heap file | Enabled |
|--------------------|---------|

**\* User modified value**

## 4. System Configuration

### 4.1. GPIO configuration

| IP    | Pin                      | Signal               | GPIO mode                    | GPIO pull/up pull down      | Max Speed | User Label |
|-------|--------------------------|----------------------|------------------------------|-----------------------------|-----------|------------|
| ADC1  | PC0                      | ADC1_INP10           | Analog mode                  | No pull-up and no pull-down | n/a       |            |
|       | PC1                      | ADC1_INP11           | Analog mode                  | No pull-up and no pull-down | n/a       |            |
|       | PC2                      | ADC1_INP12           | Analog mode                  | No pull-up and no pull-down | n/a       |            |
|       | PC3                      | ADC1_INP13           | Analog mode                  | No pull-up and no pull-down | n/a       |            |
|       | PA2                      | ADC1_INP14           | Analog mode                  | No pull-up and no pull-down | n/a       |            |
|       | PA3                      | ADC1_INP15           | Analog mode                  | No pull-up and no pull-down | n/a       |            |
|       | PB1                      | ADC1_INP5            | Analog mode                  | No pull-up and no pull-down | n/a       |            |
| DAC1  | PA4                      | DAC1_OUT1            | Analog mode                  | No pull-up and no pull-down | n/a       |            |
|       | PA5                      | DAC1_OUT2            | Analog mode                  | No pull-up and no pull-down | n/a       |            |
| DEBUG | PA13(JTMS/<br>SWDIO)     | DEBUG_JTMS-<br>SWDIO | n/a                          | n/a                         | n/a       |            |
|       | PA14(JTCK/<br>SWCLK)     | DEBUG_JTCK-<br>SWCLK | n/a                          | n/a                         | n/a       |            |
| RCC   | PH0-<br>OSC_IN(PH<br>0)  | RCC_OSC_IN           | n/a                          | n/a                         | n/a       |            |
|       | PH1-<br>OSC_OUT(P<br>H1) | RCC_OSC_OUT          | n/a                          | n/a                         | n/a       |            |
| SPI2  | PB13                     | SPI2_SCK             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|       | PB14                     | SPI2_MISO            | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|       | PB15                     | SPI2_MOSI            | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
| TIM1  | PA8                      | TIM1_CH1             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|       | PA9                      | TIM1_CH2             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
| TIM2  | PA15(JTDI)               | TIM2_CH1             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|       | PB3(JTDO/T<br>RACESWO)   | TIM2_CH2             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
| TIM3  | PA6                      | TIM3_CH1             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|       | PA7                      | TIM3_CH2             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
| TIM4  | PB6                      | TIM4_CH1             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|       | PB7                      | TIM4_CH2             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
| TIM5  | PA0                      | TIM5_CH1             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|       | PA1                      | TIM5_CH2             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
| TIM8  | PC6                      | TIM8_CH1             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|       | PC7                      | TIM8_CH2             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|       | PC8                      | TIM8_CH3             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
|       | PC9                      | TIM8_CH4             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |
| UART4 | PA11                     | UART4_RX             | Alternate Function Push Pull | No pull-up and no pull-down | Low       |            |

| IP   | Pin                             | Signal      | GPIO mode                                                     | GPIO pull/up pull down      | Max Speed | User Label |
|------|---------------------------------|-------------|---------------------------------------------------------------|-----------------------------|-----------|------------|
|      | PA12                            | UART4_TX    | Alternate Function Push Pull                                  | No pull-up and no pull-down | Low       |            |
| GPIO | PC13                            | GPIO_Output | Output Push Pull                                              | No pull-up and no pull-down | Low       |            |
|      | PC14-<br>OSC32_IN(<br>OSC32_IN) | GPIO_Output | Output Push Pull                                              | No pull-up and no pull-down | Low       |            |
|      | PC4                             | GPIO_Output | Output Push Pull                                              | No pull-up and no pull-down | Low       |            |
|      | PC5                             | GPIO_Output | Output Push Pull                                              | No pull-up and no pull-down | Low       |            |
|      | PB0                             | GPIO_Input  | Input mode                                                    | No pull-up and no pull-down | n/a       |            |
|      | PB2                             | GPIO_Output | Output Push Pull                                              | No pull-up and no pull-down | Low       |            |
|      | PB10                            | GPIO_EXTI10 | External Interrupt Mode with<br>Rising edge trigger detection | <b>Pull-up *</b>            | n/a       |            |
|      | PB12                            | GPIO_Output | Output Push Pull                                              | No pull-up and no pull-down | Low       |            |
|      | PC10                            | GPIO_Input  | Input mode                                                    | No pull-up and no pull-down | n/a       |            |
|      | PC11                            | GPIO_Output | Output Push Pull                                              | No pull-up and no pull-down | Low       |            |
|      | PC12                            | GPIO_Output | Output Push Pull                                              | No pull-up and no pull-down | Low       |            |
|      | PD2                             | GPIO_Output | Output Push Pull                                              | No pull-up and no pull-down | Low       |            |

## 4.2. GPDMA1

**Channel 2 - 2 Words Internal FIFO**

**: Standard Request Mode**

**Channel 1 - 2 Words Internal FIFO**

**: Standard Request Mode**

**Channel 0 - 2 Words Internal FIFO**

**: Standard Request Mode**

### 4.2.1. All Channels:

#### **Channel 0:**

Request

**ADC1 \***

#### **Channel 1:**

Request

**SPI2\_TX \***

#### **Channel 2:**

Request

**SPI2\_RX \***

### 4.2.2. SECURITY:

#### **CH2:**

Enable Channel as Privileged

NON PRIVILEGED

#### **CH1:**

Enable Channel as Privileged

NON PRIVILEGED

#### **CH0:**

Enable Channel as Privileged

NON PRIVILEGED

#### 4.2.3. CH2:

##### **Circular configuration:**

Circular Mode

Disable

##### **Request Configuration:**

Request

**SPI2\_RX/I2S2\_RX \***

DMA Handle in IP Structure

hdmarx

Block HW request protocol

Single/Burst Level

##### **Channel configuration:**

Priority

**Medium \***

Transaction Mode

Normal

Direction

Peripheral To Memory

##### **Source Data Setting:**

Source Address Increment After Transfer

Disabled

Data Width

Byte

Burst Length

1

Allocated Port for Transfer

Port 0

##### **Destination Data Setting:**

Destination Address Increment After Transfer

**Enabled \***

Data Width

Byte

Burst Length

1

Allocated Port for Transfer

Port 0

##### **Data Handling:**

Data Handling Configuration

Disable

##### **Trigger:**

Trigger Configuration

Disable

##### **Transfer Event Configuration:**

Transfer Event Generation

The TC (and the HT) event is generated at the (respectively half) end of each block

#### 4.2.4. CH1:

##### **Circular configuration:**

Circular Mode

Disable

##### **Request Configuration:**

Request

**SPI2\_TX/I2S2\_TX \***

DMA Handle in IP Structure

hdmatx

Block HW request protocol

Single/Burst Level

#### Channel configuration:

|                  |                               |
|------------------|-------------------------------|
| Priority         | <b>Medium *</b>               |
| Transaction Mode | Normal                        |
| Direction        | <b>Memory To Peripheral *</b> |

#### Source Data Setting:

|                                         |                  |
|-----------------------------------------|------------------|
| Source Address Increment After Transfer | <b>Enabled *</b> |
| Data Width                              | Byte             |
| Burst Length                            | 1                |
| Allocated Port for Transfer             | Port 0           |

#### Destination Data Setting:

|                                              |          |
|----------------------------------------------|----------|
| Destination Address Increment After Transfer | Disabled |
| Data Width                                   | Byte     |
| Burst Length                                 | 1        |
| Allocated Port for Transfer                  | Port 0   |

#### Data Handling:

|                             |         |
|-----------------------------|---------|
| Data Handling Configuration | Disable |
|-----------------------------|---------|

#### Trigger:

|                       |         |
|-----------------------|---------|
| Trigger Configuration | Disable |
|-----------------------|---------|

#### Transfer Event Configuration:

|                           |                                                                                     |
|---------------------------|-------------------------------------------------------------------------------------|
| Transfer Event Generation | The TC (and the HT) event is generated at the (respectively half) end of each block |
|---------------------------|-------------------------------------------------------------------------------------|

### 4.2.5. CH0:

#### Circular configuration:

|               |                 |
|---------------|-----------------|
| Circular Mode | <b>Enable *</b> |
| Circular Port | Port 0          |

#### Request Configuration:

|                            |                    |
|----------------------------|--------------------|
| Request                    | <b>ADC1 *</b>      |
| DMA Handle in IP Structure | DMA_Handle         |
| Block HW request protocol  | Single/Burst Level |

#### Channel configuration:

|                  |                         |
|------------------|-------------------------|
| Priority         | <b>High *</b>           |
| Transaction Mode | Normal                  |
| Direction        | Peripheral To Memory    |
| Node Type        | GPDMA Linear Addressing |

#### Source Data Setting:

|                                         |                    |
|-----------------------------------------|--------------------|
| Source Address Increment After Transfer | Disabled           |
| Data Width                              | <b>Half Word *</b> |
| Burst Length                            | 1                  |

|                                                     |                                                                                     |
|-----------------------------------------------------|-------------------------------------------------------------------------------------|
| Allocated Port for Transfer                         | Port 0                                                                              |
| <b>Destination Data Setting:</b>                    |                                                                                     |
| Destination Address Increment After Transfer        | <b>Enabled *</b>                                                                    |
| Data Width                                          | <b>Half Word *</b>                                                                  |
| Burst Length                                        | 1                                                                                   |
| Allocated Port for Transfer                         | Port 0                                                                              |
| <b>Data Handling:</b>                               |                                                                                     |
| Data Handling Configuration                         | Disable                                                                             |
| <b>Trigger:</b>                                     |                                                                                     |
| Trigger Configuration                               | Disable                                                                             |
| <b>Transfer Event Configuration:</b>                |                                                                                     |
| Transfer Event Generation                           | The TC (and the HT) event is generated at the (respectively half) end of each block |
| <b>Channel Configuration for Linked List:</b>       |                                                                                     |
| Execution Mode (circular/linear) of the Linked List | Circular                                                                            |
| Linked List Execution Mode                          | The List is fully executed                                                          |

4.3. GPDMA2

4.4. LINKEDLIST

## 4.5. NVIC configuration

### 4.5.1. NVIC

| Interrupt Table                         | Enable | Preenmption Priority | SubPriority |
|-----------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                  | true   | 0                    | 0           |
| Hard fault interrupt                    | true   | 0                    | 0           |
| Memory management fault                 | true   | 0                    | 0           |
| Pre-fetch fault, memory access fault    | true   | 0                    | 0           |
| Undefined instruction or illegal state  | true   | 0                    | 0           |
| System service call via SWI instruction | true   | 0                    | 0           |
| Debug monitor                           | true   | 0                    | 0           |
| Pendable request for system service     | true   | 15                   | 0           |
| System tick timer                       | true   | 15                   | 0           |
| GPDMA1 Channel 0 global interrupt       | true   | 5                    | 0           |
| GPDMA1 Channel 1 global interrupt       | true   | 5                    | 0           |
| GPDMA1 Channel 2 global interrupt       | true   | 5                    | 0           |
| SPI2 global interrupt                   | true   | 5                    | 0           |
| TIM17 global interrupt                  | true   | 15                   | 0           |
| Flash non-secure global interrupt       | unused |                      |             |
| RCC non-secure global interrupt         | unused |                      |             |
| EXTI Line10 interrupt                   | unused |                      |             |
| ADC1 global interrupt                   | unused |                      |             |
| DAC1 interrupt                          | unused |                      |             |
| TIM1 Break interrupt                    | unused |                      |             |
| TIM1 Update interrupt                   | unused |                      |             |
| TIM1 Trigger and Commutation interrupts | unused |                      |             |
| TIM1 Capture Compare interrupt          | unused |                      |             |
| TIM2 global interrupt                   | unused |                      |             |
| TIM3 global interrupt                   | unused |                      |             |
| TIM4 global interrupt                   | unused |                      |             |
| TIM5 global interrupt                   | unused |                      |             |
| UART4 global interrupt                  | unused |                      |             |
| TIM8 Break interrupt                    | unused |                      |             |
| TIM8 Update interrupt                   | unused |                      |             |
| TIM8 Trigger and Commutation interrupts | unused |                      |             |
| TIM8 Capture Compare interrupt          | unused |                      |             |
| FPU global interrupt                    | unused |                      |             |
| Instruction cache global interrupt      | unused |                      |             |

### 4.5.2. NVIC Code generation

|                         |                 |              |                  |
|-------------------------|-----------------|--------------|------------------|
| Enabled interrupt Table | Select for init | Generate IRQ | Call HAL handler |
|-------------------------|-----------------|--------------|------------------|

| Enabled interrupt Table                 | Select for init<br>sequence ordering | Generate IRQ<br>handler | Call HAL handler |
|-----------------------------------------|--------------------------------------|-------------------------|------------------|
| Non maskable interrupt                  | false                                | true                    | false            |
| Hard fault interrupt                    | false                                | true                    | false            |
| Memory management fault                 | false                                | true                    | false            |
| Pre-fetch fault, memory access fault    | false                                | true                    | false            |
| Undefined instruction or illegal state  | false                                | true                    | false            |
| System service call via SWI instruction | false                                | false                   | false            |
| Debug monitor                           | false                                | true                    | false            |
| Pendable request for system service     | false                                | false                   | false            |
| System tick timer                       | false                                | false                   | true             |
| GPDMA1 Channel 0 global interrupt       | false                                | true                    | true             |
| GPDMA1 Channel 1 global interrupt       | false                                | true                    | true             |
| GPDMA1 Channel 2 global interrupt       | false                                | true                    | true             |
| SPI2 global interrupt                   | false                                | true                    | true             |
| TIM17 global interrupt                  | false                                | true                    | true             |

\* User modified value



## 5. System Views

### 5.1. Category view

#### 5.1.1. Current

Middleware

Software Packs

X-CUBE-FREERTOS ✓

| System Core | Analog | Timers | Connectivity | Multimedia | Security | Computing | Trace and Debug | Power and Thermal | Utilities  | Other |
|-------------|--------|--------|--------------|------------|----------|-----------|-----------------|-------------------|------------|-------|
| CORTEX_M33  | ADC1 ✓ | TIM1 ✓ | SPI2 ✓       |            |          |           | DEBUG ✓         | PWR ✓             | LINKEDLIST |       |
| GPDMA1 ✓    | DAC1 ✓ | TIM2 ✓ | UART4 ✓      |            |          |           |                 |                   |            |       |
| GPDMA2      |        | TIM3 ✓ |              |            |          |           |                 |                   |            |       |
| GPIO ✓      |        | TIM4 ✓ |              |            |          |           |                 |                   |            |       |
| ICACHE ✓    |        | TIM5 ✓ |              |            |          |           |                 |                   |            |       |
| NVIC ✓      |        | TIM8 ✓ |              |            |          |           |                 |                   |            |       |
| RCC ✓       |        |        |              |            |          |           |                 |                   |            |       |
| SYS ✓       |        |        |              |            |          |           |                 |                   |            |       |

## 6. Software Pack Report

### 6.1. Software Pack selected

| Vendor             | Name            | Version | Component                                                                                                                                                                                           |
|--------------------|-----------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STMicroelectronics | X-CUBE-FREERTOS | 1.3.1   | Class : CMSIS<br>Group : RTOS2<br>SubGroup : Core<br>Variant :<br>TZ_Non_Supported<br>Version : 10.6.2<br>Class : CMSIS<br>Group : RTOS2<br>SubGroup : Heap<br>Variant : Heap_1<br>Version : 10.6.2 |

## 7. Docs & Resources

| Type                    | Link                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BSDL files              | <a href="https://www.st.com/resource/en/bsdl_model/stm32h5-bsdl.zip">https://www.st.com/resource/en/bsdl_model/stm32h5-bsdl.zip</a>                                                                                                                                                                                                                                                                                                                                 |
| IBIS models             | <a href="https://www.st.com/resource/en/ibis_model/stm32h5-ibis.zip">https://www.st.com/resource/en/ibis_model/stm32h5-ibis.zip</a>                                                                                                                                                                                                                                                                                                                                 |
| System View Description | <a href="https://www.st.com/resource/en/svd/stm32h5-svd.zip">https://www.st.com/resource/en/svd/stm32h5-svd.zip</a>                                                                                                                                                                                                                                                                                                                                                 |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf</a>                                                                                                                                                                                                                                                         |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf">https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf</a>                                                                                                                                                                                                                                                                                 |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf</a>                                                                                                                                                                                                                                                           |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h5-series-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h5-series-overview.pdf</a>                                                                                                                                                                                                                                                     |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf</a>                                                                                                                                                                                                                                                         |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-entry-level-graphics.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-entry-level-graphics.pdf</a>                                                                                                                                                                                                                                               |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf">https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf</a>                                                                                                                                                                                                                                                                 |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf">https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf</a>                                                                                                                                                                                                                                                               |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32nucleo.pdf">https://www.st.com/resource/en/flyer/flstm32nucleo.pdf</a>                                                                                                                                                                                                                                                                                                                                         |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32h5.pdf">https://www.st.com/resource/en/flyer/flstm32h5.pdf</a>                                                                                                                                                                                                                                                                                                                                                 |
| Security Bulletin       | <a href="https://www.st.com/resource/en/technical_note/tn1474-security-bulletin-tn1474stpsirt-information-on-softwarebased--microarchitectural-timing-sidechannel-attacks-on-mcus-with-trustzone-for--armv8m-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1474-security-bulletin-tn1474stpsirt-information-on-softwarebased--microarchitectural-timing-sidechannel-attacks-on-mcus-with-trustzone-for--armv8m-stmicroelectronics.pdf</a> |
| Security Bulletin       | <a href="https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf</a>                                                                                                                             |

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| Security Bulletin | <a href="https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf">https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf</a>                                                                   |
| Application Notes | <a href="https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf</a>                                                                                                                   |
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| Application Notes | <a href="https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf</a>                                                                                                                       |
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for related Tools  
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Application Notes [https://www.st.com/resource/en/application\\_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf)  
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& Software

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for related Tools  
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& Software

Application Notes [https://www.st.com/resource/en/application\\_note/an4502-stm32-](https://www.st.com/resource/en/application_note/an4502-stm32-)



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| for related Tools & Software                   | <a href="#">smbuspmibus-expansion-package-for-stm32cube-stmicroelectronics.pdf</a>                                                                                                                                                                                                                                        |
| Application Notes for related Tools & Software | <a href="https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf</a>               |
| Application Notes for related Tools & Software | <a href="https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf</a>                                                                               |
| Application Notes for related Tools & Software | <a href="https://www.st.com/resource/en/application_note/an6007-getting-started-with-stirot-st-immutable-root-of-trust-for-stm32h5-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6007-getting-started-with-stirot-st-immutable-root-of-trust-for-stm32h5-mcus-stmicroelectronics.pdf</a> |
| Application Notes for related Tools & Software | <a href="https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf</a>                         |
| Application Notes for related Tools & Software | <a href="https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf</a>                                   |
| Application Notes for related Tools & Software | <a href="https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf</a>                                             |
| Application Notes for related Tools & Software | <a href="https://www.st.com/resource/en/application_note/an6382-introduction-to-swd-multidrop-for-stm32-with-stlink-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6382-introduction-to-swd-multidrop-for-stm32-with-stlink-stmicroelectronics.pdf</a>                                         |
| Errata Sheets                                  | <a href="https://www.st.com/resource/en/errata_sheet/es0565-stm32h562xx563xx573xx-device-errata-stmicroelectronics.pdf">https://www.st.com/resource/en/errata_sheet/es0565-stm32h562xx563xx573xx-device-errata-stmicroelectronics.pdf</a>                                                                                 |
| Datasheet                                      | <a href="https://www.st.com/resource/en/datasheet/dm00956662.pdf">https://www.st.com/resource/en/datasheet/dm00956662.pdf</a>                                                                                                                                                                                             |
| Programming Manuals                            | <a href="https://www.st.com/resource/en/programming_manual/pm0264-stm32-cortexm33-mcus-and-mpus-programming-manual-stmicroelectronics.pdf">https://www.st.com/resource/en/programming_manual/pm0264-stm32-cortexm33-mcus-and-mpus-programming-manual-stmicroelectronics.pdf</a>                                           |
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| Technical Notes & Articles                     | <a href="https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf</a> |

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| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf</a>                                         |
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| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf</a>                       |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf</a>                         |
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| User Manuals                  | <a href="https://www.st.com/resource/en/user_manual/um3150-stm32h5-series-safety-manual-stmicroelectronics.pdf">https://www.st.com/resource/en/user_manual/um3150-stm32h5-series-safety-manual-stmicroelectronics.pdf</a>                                                                                                                                     |
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